MIL-W-48272 (AR)
Amendment 8
28 October 1980
SUPERSEDING
Amendment 7
10 February 1980

MILITARY SPECIFICATION

WARHEAD SECTION, GM, HE: XM251
LOADING, ASSEMBLING AND PACKING

This Amendment forms a part of MIlitary Specification MIL-W-48272(AR), dated 26 November 1973.

PAGE 1

Title - Add "(M251), M251E1"

Para 1.1, Delete in its entirety and replace with the following:

1.1 This specification covers the loading, assembling and packing for two types of warhead sections designated as Warhead Sections, GM HE: XM251 (M251), and M251E1.

Para 2.1, Specifications, Military - Add the following:

DOD-D-1000 -Drawing, Engineering and Associated List

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Para 2.1, Standards, Military - Add the following:

DOD-STD-100 -Engineering Drawing Practices
MIL-STD-1167 -Ammunition Data Card

Para 2.1, Add the following:

HANDBOOKS

MILITARY

MIL-HDBK-204 - Inspection Equipment Design

FSC: 1340

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Para 2.1, Drawings, Change "U.S. Army Armament Command" to "U.S. Army Armament Research and Development Command (ARRADCOM) Product and Packaging Drawings".

Para 2.1, Drawings, U.S. Army Armament Command, Add:

	9211190	-Pkg and Mkg for Warhead Sect., GM, HE: XM251; Prac: XM198E1; Tng: XM201
	9200664	-Universal C. G. and Weight Gage
*	9208657	-Cable Assembly, M251
	9266382	-Roll Moment Fixture
	9266388	-Equipment for Radial C. G. and Principal Axis Inclination
	9266385	-Adapter
	9266387	-Adapter
	9266418	-Test Fixture, Insulation Resistance
	9266419	-Test Procedure and Calibration Instr. Insulation Resistance Test for Warhead Section Cable Assembly - XM251 for C9266418
	9266422	-Continuity and Resistance, LANCE Warhead with Fuze
	9266424	-Weight Measurement Fixture on Lance Warhead XM251
	9266425	-Operating Instructions and Calibration Procedures for Weight Measurement on Lance Warhead XM251 for B9266624
	9266426	-Pitch and Yaw Moment of Inertia Fixture
	9266386	-Adapter
	9266427	-Operating Instructions Lance Warhead, XM251 (to be used with A9266426)
	9266428	-Operating Instructions Lance Warhead, XM251 (to be used with F9266382)
	9266429	-Operating Procedures for 9266383 XM251 Lance Warhead
	9266430	-Operating Instructions XM251 Lance Warhead (to be used with F9200664)
*	9283999	-Metal Parts Assembly M251
	9332549	-Warhead Section, GM, HE: M251E1

Para 2.1, Drawings, US Army Armament Command - Change "XM251" to "M251".

Para 2.1, Drawings US Army Armament Command - Add:

	9266435	-Test Set, Continuity and Resistance Warhead Section M251

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9266439 -Calibration Procedures for Dynamic Balance Machine, XM251 Lance Warhead

Para 2.1, Drawings, US Army Armament Command, Dwg. 9266429 - Change "9266383" to "9266388".

Para 2.1, Drawings, US Army Armament Command, Dwg. 9266425 - Change "B9266624" to "9266424".

Para 2.1, Drawings, U.S. Army Armament Command - Delete the following drawings in their entirety: 9200664, 9266388, 9266385, 9266387, 9266386, 9266429, 9266430 and 9266439.

Para 2.1, Drawings, U.S. Army Armament Command - Add: "M251E1" to the title of the following drawings: 9211190, 9266419, 9266424, 9266425, 9266427, 9266428, and, 9266435.

* Para 2.1, Drawings - Add:

US Army Missile Command

11430220 -Gage Assembly, Acceptance Warhead Interface, M.S. 100.00

11430988 -Instruction Tag, Warhead Gage for Missile Interface

Para 2.2 - Add the following:

"AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D1621-73 -Test for Compressive Strength of Rigid Cellular Plastics

ASTM D1622-63 -Test for Apparent Density of Rigid Cellular Plastics

Para 3.2, - Change "..on Drawing F-9284000," to "...on Drawings 9284000 and 9332549,".

Para 3.4 - Delete in its entirety.

Para 3.4.a - Delete in its entirety.

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Para 3.4.b - Add the following:

$\underline{C}$ (Pos)	M	0.5 Max.
$\underline{C}$ (Pos)	H	3.0K $\pm$ 1.2K
$\underline{C}$ (Pos)	G	5.0K $\pm$ 2.0K
$\overline{P}$	C	50 Megohms Min

Para 3.4.b - Delete in its entirety.

Para 3.5 - Delete in its entirety.

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Para 4.3.2.1, Title - Change "XM251" to "M251".

* Para 4.3.2.1, Major Defects - Change "100% Inspection" to "AQL = 0.40% (Note 1)"

Para 4.3.2.1 - Delete Major 106 and substitute:

111. Distance from aft ring assembly to longitudinal end primer (with booster cover off) - See View L

Para 4.3.2.1, Major 107 - Change "barrier cover" to "booster cover".

Para 4.3.2.1 - Add Note 1 as follows:

"Note 1. For Lot Sizes below 90 items, the AQL shall equal 1.0%"

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Para 4.3.2.1 - Minor 201 - Change Code No. "01010" to "01012".

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Para 4.3.2.7 - Add the following:

104. Holes missing Visual 07005

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Para 4.3.2.10 - Delete in its entirety and replace with the following:

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4.3.2.10 Warhead Section, during loading, GM, HE: M251 and M251E1 (See Dwgs. 9284000 and 9332549).

Categories		Defects	Method of Inspection	Code No.
Critical: None defined				
		Major:	100% Inspection	
	101.	Presence of plate (Dwg. 9331775) prior to loading. (M251E1 only)	Visual	10013
	102.	Presence of aft filler (M251E1 only)	Visual	10014
	103.	Containment block pad missing or improperly installed	Visual	10001
	104.	Wood blocks missing	Visual	10002
	105.	Distance to bombs and wood blocks	Gage	10003
	106.	Polyethlene sheets improperly installed, torn, or missing	Visual	10004
	107.	Incorrect number of bag and bag filler assemblies	Visual	10005
*	108.	Bag and bag filler assembly flaps facing forward	Visual	10006
	109.	Wedges missing or improperly assembled	Visual	10007
	110.	Skin missing or improperly assembled	Visual	10008
	111.	Spacer missing or improperly assembled	Visual	10009
	112.	Insulation tape missing damaged or improperly installed on aft ring assembly	Visual	10010
	113.	O-Ring missing, damaged or improperly installed on aft ring assembly	Visual	10011
Minor:				
	201.	Evidence of poor workmanship (see 3.6)	Visual	10012

* Para 4.3.2.11, Title - Add "and M251E1"

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Para 4.3.2.11, Title - Change "(See Dwg. 9384000)" to "(See Dwgs. 9284000 and 9332549)".

Para 4.3.2.11 - Change "XM251" to "M251".

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Para 4.3.2.11 - Major 105 - Delete "See Dwg. 9283999 - View L" and substitute "See Section G-G".

Paragraph 4.3.2.11 - Add the following:

114	Shims or adhesive covering ends of booster that is inserted through booster cover hole (prior to assembly of booster)	Visual	11015
115	Shims or adhesive on side of booster that contacts end primer in MDF and primer assembly (after assembly of booster)	Visual	11016

Paragraph 4.3.2.11 - Add the following:

116	Aft skin above aft ring assembly	Gage	11017

Para 4.3.2.11 Major 106 - Delete in its entirety and substitute:

117	Distance from aft ring assembly to longitudinal end primer (with booster cover off) - See Section G-G		11018

Para 4.3.2.11, Major 116 - Delete in its entirety.

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Para 4.3.2.12, Title - Add "and M251E1"

Para 4.3.2.12, Title - Change "(See Dwg. 9284000)" to "(See Dwgs. 9284000 and 9332549)".

Paragraph 4.3.2.12 - Change "XM251" to M251".

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Paragraph 4.3.2.12 - Add the following:

110	Booster (3) installed in reversed direction.	Visual	12012

Para 4.3.2.13 - Change "XM251" to "M251".

Para 4.3.2.13, Title - Add "and M251E1".

Para 4.3.2.13, Title - Change "(See Dwg. 9284000)" to "(See Dwgs. 9284000 and 9332549)".

Para 4.3.2.13, Major 101 - Delete phase "and between forward and aft skins".

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Para 4.3.2.13, Major 104 - Delete in its entirety and substitute:

104	Tee bolts bind or too loose in cavities	Manual	13004

Para 4.3.2.13, Major 106 - Delete "Gage" and substitute "Dwgs. 9266424 and 9266425".

Para 4.3.2.13, Major 107 - Delete "Test" and substitute "Dwgs. 9200664, 9266387 and 9266430".

Para 4.3.2.13, Major 109 - Delete "Test" and substitute "Dwgs. 9266426 and 9266427".

Para 4.3.2.13, Major 110 - Delete "Test" and substitute "Dwgs. 9266382 and 9266428".

Para 4.3.2.13, Major 112 - Change "Gage" to "Visual".

Para 4.3.2.13, - Add the following:

115.	Forward facing step at nose cone assembly, max.	Gage	13016
116.	Aft facing step at nose cone assembly, max.	Gage	13017

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Para 4.3.2.13 - Add the following:

117.	Marking missing, incomplete or illegible.	Visual	13018

Para 4.3.2.13, Major 107 - Delete "Dwgs. 9200664, 9266387 and 9266430" and substitute "Test".

* Para 4.3.2.13, Major 111 - Delete "Gage" and substitute "11430220/11430988".

Para 4.3.2.14, Title - Add "and M251E1"

Para 4.3.2.14, Title - Change "XM251" to "M251".

Para 4.3.2.14 - Add "100% Inspection" to Majors.

Para 4.3.2.15, Title - Add "and M251E1".

Para 4.3.2.15 - Title - Change "XM251" to "M251".

Para 4.3.2.15, Major Defects - Add: "100% Inspection".

* Para 4.3.2.15 - Add Major 106 as follows:

"106.	Stacking bolts, missing or damaged.	Visual	15007"

Add the following:

4.3.2.16 Spacer, foam (See Dwg. 9312646). -

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major			
101.	Thickness	Gage	23001
102.	Depth of cut (2 sides)	Gage	23002

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Minor

201.	Outside diameter	Gage	23003
202.	Inside diameter	Gage	23004
203.	Width of small cutout	Gage	23005
204.	Depth of small cutout	Gage	23006
205.	Width of large cutout	Gage	23007
206.	Depth of large cutout	Gage	23008
207.	Evidence of poor workmanship (See 3.6)	Visual	23009

Add Para: 4.3.2.17 as follows:

4.3.2.17 Filler, Aft (See Dwg. 9331772)

Categories	Defects	Method of Inspection	Code No.

Critical: None defined

Major:

101.	Length	Gage	24001
102.	Largest outside diameter	Gage	24002
103.	Second largest outside diameter	Gage	24003
104.	Inside diameter	Gage	24004
105.	Density (Note 1)	Test	24005
106.	Compressive strength (Note 1)	Test	24006
107.	Surface Finish (See Note 2)	Visual	24007
108.	Forward lead angle improper	Gage	24008

Minor:

201.	Evidence of poor workmanship (See 3.6)	Visual	24009

Note 1: A three (3) sample (from 1 part) average shall be used to verify compliance.

Note 2: The surface finish shall be smooth and hard (as molded) with no evidence of loose, crumbling particles.

Add para 4.3.2.18 as follows:

4.3.2.18 Filler, Nose (See Dwg. 9331774)

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Length	Gage	25001
102.	Largest outside diameter	Gage	25002
103.	Second largest outside diameter	Gage	25003
104.	Inside diameter	Gage	25004
105.	Width of small cutout (2 places)	Gage	25005
106.	Width of large cutout (2 places)	Gage	25006
107.	Depth of small cutout (2 places)	Gage	25007
108.	Depth of large cutout (2 places)	Gage	25008
109.	Chamfer	Visual	25009
110.	Density (Note 1)	Test	25010
111.	Compressive strength (Note 1)	Test	25011
Minor:			
201.	Evidence of poor workmanship (See 3.6).	Visual	25010

Note 1: A three (3) sample (from 1 part) average shall be used to verify compliance.

Add Para 4.3.2.19 as follows:

4.3.2.19 Plate (See Dwg. 9331775)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:			
101.	Outside diameter	Gage	26001
102.	Inside diameter	Gage	26002
103.	Thickness	Gage	26003
104.	Protective finish	Visual	26004

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Minor:

201. Evidence of poor workmanship (see 3.6) 26005

Para 4.3.3.2 - Add after "4.3.4", "4.3.4", "(See Dwgs. 9266418 and 9266419)".

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Para 4.3.3.3 - Add after "4.3.4", "(See Dwg. 9266435)"

Para 4.3.3.4 - Add after "4.3.4", "(See Dwg. 9266418 and 9266419). This test to be performed after performance of tests and inspections in Para 4.3.2.13, Majors 106 thru 110 and in Para 4.3.3.7."

Para 4.3.3.4 - Delete the phase - "and in Para 4.3.3.7".

Para 4.3.3.5 - Add after second sentence, "This test to be performed after performance of tests and inspections in Para 4.3.2.13, Majors 106 thru 110 and in Para 4.3.3.7.

Para 4.3.3.5 - Delete the phrase - "and in Para 4.3.3.7".

Para 4.3.3.5 - Add after "4.3.4", "(See Dwg 9266435)".

Para 4.3.3.6 - Add after "4.3.4", "(See Dwg. 9266422 for equipment to test for compliance with requirement in Para 3.4.b)". This test to be performed after performance of tests and inspections in Para 4.3.2.13, Majors 106 thru 110 and in Para 4.3.3.7.

Para 4.3.3.6 - Delete the phrase - "and in Para 4.3.3.7".

* Para 4.3.3.6 - Delete the phrase - "for equipment to test for compliance with requirement in Para 3.4.b".

Para 4.3.3.6 - Change "See 3.4" to "(See Dwg. 9284000)".

Para 4.3.3.6 - Change "(See Dwg. 9284000) "to "(See Dwgs. 9284000 and 9332549)".

Para 4.3.3.7 - Add after "4.3.4", "(See Dwgs 9266383, 9266385, 9266386 and 9266429)".

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Para 4.3.3.7 - Change "9266383" to "9266388".

Para 4.3.3.7 - After "9266429", add "9266439".

Para 4.3.3.7 - Delete in its entirety.

Add Para 4.3.3.8 as follows:

4.3.3.8 Density and compressive strength of polyurethane fillers (See Dwgs. 9331772 and 9331774), Major Defect, Codes 24005/25010 and 24006/25011. One (1) part from each lot of nose and aft fillers shall be selected for test. The density tests (non-destructive) shall be conducted on the compressive test specimens prior to performing the compressive tests. The compressive strength specimens shall be taken from the part after removal of the filler part skin. The three 2-inch cube specimens shall be cut at 120° intervals around the part; one at the top, one at the bottom, and one midway up the part. These tests shall be performed using equipment in accordance with 4.3.4. Any lot which fails to comply with the specified requirement shall be rejected.

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* Para 4.4.6 - Delete in its entirety and replace with the following:

4.4.6 Continuity and resistance of cable assembly of loaded warhead when fuze is installed at the time of loading. - The continuity and resistance of the cable assembly/fuze shall be tested in accordance with Dwg. 9284000 (M251) or Dwg. 9332549 (M251E1), using approved test equipment. This is a non-destructive test. Assemblies subjected to this test shall be returned to the lot.

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Para 4.4.7 - Delete in its entirety.

Add Para 4.4.9 as follows:

4.4.9 Density and compressive strength of polyurethane fillers. The density tests shall be conducted in accordance with ASTM standard D-1622 and the compressive strength tests in accordance with ASTM standard D-1621. Parts from which specimens are taken shall not be returned to the lot.

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* Add Para 4.4.10 as follows:

4.4.10 Longitudinal Center of Gravity. - The loaded warhead shall be subjected to the longitudinal center of gravity test using approved test equipment. This is a non-destructive test. Assemblies subjected to this test shall be returned to the lot.

* Add Para 4.4.11 as follows:

4.4.11 Pitch and Yaw Moment of Inertia. - The loaded warhead shall be subjected to the pitch and yaw moment of inertia test using approved test equipment. This is a non-destructive test. Assemblies subjected to this test shall be returned to the lot.

* Add Para 4.4.12 as follows:

4.4.12 Roll Moment of Inertia. - The loaded warhead shall be subjected to the roll moment of inertia test using approved test equipment. This is a non-destructive test. Assemblies subjected to this test shall be returned to the lot.

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Para 6.3.1, section b. & c. - Change "MIL-STD-100" to "DOD-STD-100".

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Para 6.3.2 - Change "(MIL-D-1000)" to "(DOD-D-1000)".

Para 6.3.3 - Change "Picatinny Arsenal design..." to "Design ...".

Para 6.4 - Delete the phase "... and ammunition data cards...".

Para 6.4 - Change forwarding address to read as follows: "Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-R, Dover, NJ 07801.

Para 6.5 - Change forwarding address to read as follows: "Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-R, Dover, NJ 07801.

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Para 6.6 - Change forwarding address to read as follows: "Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-R, Dover, NJ 07801.

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Figure 1. - Delete in its entirety.

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